

From a cited implication graph to one buildable intervention: choosing an air-pollution problem and its solution for Abu Dhabi by near-decomposability, exposure load and multi-criteria screening

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Abstract

Picking one air-pollution problem in Abu Dhabi and one intervention a student group can pilot in a semester is a selection problem, and we treat it as one. An implication graph of the city's air-pollution system was built in the WHO DPSEEA frame from 418 cited edge rows (160 nodes, 337 distinct edges, 173 DOIs, every source fetched at origin). Simon and Ando's near-decomposability, operationalised as a leakage ratio on blocks found by directed modularity and tested under 200 weight perturbations, five resolutions, two other partition algorithms and 50 seeds, returns four separable problem modules. An exposure-load criterion (people $\times$ hours $\times$ PM_{2.5} increment) ranks them with every input sourced: construction workers 16.9 million person $\mu\text{g}/\text{m}^3$ h per day, near-road residents and riders 7.0, the child commute 4.5, the school gate 0.68. Morphological generation over the modules' cited edges produced 332 interventions in four mechanism classes. Four conjunctive gates and an eleven-criterion weighted-sum rubric screened them, stochastic multicriteria acceptability analysis over the weights (10,000 draws) measured the ranking's dependence on the weights, and a failure-mode analysis in the premortem form fixed the design specification. The intervention that meets it, a per-bus cabin air service (diagnosis by a cabin-to-roof sensor ratio, a source fix kit, a climate-aware idle timer, a published per-bus grade), scores 79 of 90 and holds rank 1 in 86 % of draws with no weight information at all, in 90 % of draws centred on our weights and under every leave-one-criterion-out ranking. Its physics and hardware carry twenty years of measured evidence, none of it yet applied to the 237,111 pupils who ride 8,568 buses in Abu Dhabi.

1 The question and the method

The week 3 brief asks for four things: restate the need, show every solution explored, state the filters used including our own, and apply them to one final solution. That is a two-level selection: a problem from a system in which everything touches everything, then an intervention from the many that could act on it. Each level is answered here by a named, published method with a reproducible artefact (Figure 1). The literature harvest that produced the edge table was scripted lane by lane, every candidate source fetched at its origin before its row was accepted, and every row carries its citation, identifier and the sentence it rests on. The graph, the decomposition, the exposure arithmetic, the intervention inventory, every gate decision and every score are the authors' work, and the files that reproduce every figure are listed in Appendix A.

2 Step 1: the evidence graph

Nodes follow the DPSEEA chain the WHO adopted for environmental-health indicators (Corvalán et al. 1999): a *driving force* creates a *pressure* (an emission), which changes a *state* (a concentration at a place), to which people are *exposed* for a time, with an *effect*, and on which an *action* acts. Each node is a defined quantity with a unit where one exists and a scope. The list has 26 driving forces, 33 pressures, 34 states, 19 exposures, 20 effects and 28 actions. An edge is a directed influence documented by a fetched source and carries a sign, a weight, a tier, a scope, the citation, an identifier and the sentence it rests on (Table 1). Where an Abu Dhabi or UAE source-apportionment number exists the measured share is recorded beside the ordinal weight (29 of 418 rows). Duplicate rows for a node pair are kept as separate evidence lines and the pair takes the maximum weight.

The harvest ran in eight lanes, one per layer pair or theme, in two passes, the second adding nodes for every problem raised in class so that each candidate has comparable detail (Figure 2). A validator rejected

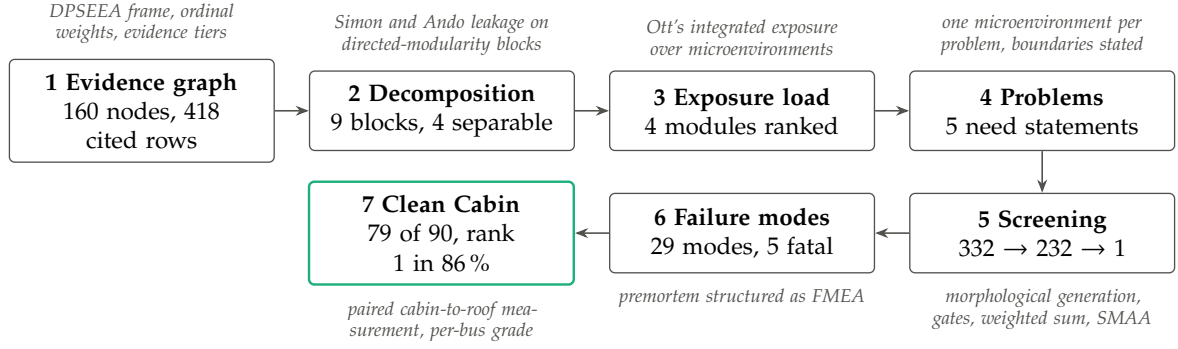


Figure 1: The seven steps and their counts. Each box is a section below, each method line names the published method the step instantiates.

Table 1: Edge rules, fixed before the harvest.

field	level	rule
weight	3	the source identifies the origin as the dominant or primary driver of the destination, or a quantified contribution of 30 % or more
	2	a substantial documented contribution, 10 to 30 %, or “major but not primary”
	1	documented and minor, under 10 %, or only the mechanism is shown
tier	1 to 4	systematic review or WHO-grade synthesis, primary measured study, modelling study or official statistic or technical report, grey literature
scope		Abu Dhabi, UAE, Gulf or global, with cross-boundary sources (regional dust, transboundary PM) kept as source nodes

any row without an identifier or a fetch flag. The result is 418 edge rows, 337 distinct edges, 238 source lines and 173 distinct DOIs, all fetched on 20 September 2026: by weight 97, 179 and 142 rows at levels 1, 2 and 3, by tier 95, 217, 78 and 28, by scope 243 global, 64 Gulf, 38 UAE and 73 Abu Dhabi. The total internal weight is 720 and five nodes are isolated. Figure 3 shows the whole graph.

3 Step 2: near-decomposability finds the problem modules

Simon and Ando (1961) showed that a system whose interaction matrix is block-diagonal apart from small off-block terms behaves, in the short run, as independent subsystems, and Simon (1962) argued that this is what makes complex systems tractable. A problem in a coupled system is therefore a block whose internal coupling dominates its coupling to the rest. For a node set S with weights $w_{ij} \in \{1, 2, 3\}$,

$$W_{\text{in}}(S) = \sum_{i,j \in S} w_{ij}, \quad \varepsilon_{\text{out}} = \frac{\sum_{i \in S, j \notin S} w_{ij}}{W_{\text{in}}}, \quad \varepsilon_{\text{in}} = \frac{\sum_{i \notin S, j \in S} w_{ij}}{W_{\text{in}}}, \quad \varepsilon_{\text{sym}} = \varepsilon_{\text{out}} + \varepsilon_{\text{in}}, \quad (1)$$

and $\varepsilon_{\text{transfer}}$ is the part of ε_{out} that lands on driver, pressure or state nodes, the load that acting inside S moves elsewhere. A block *qualifies* when it holds an action node and an exposure or effect node, and is *separable* when $\varepsilon_{\text{sym}} < 1$. An anchor test with two planted dense blocks joined by one weak edge returns a planted block with leakage equal to that edge over W_{in} , which fixes the scale. Blocks come from directed modularity (Leicht and Newman 2008) maximised by Louvain (Blondel et al. 2008) with the directed gain of Dugué and Perez (2015), resolution 1.0, seed 0.

Partition. $Q = 0.604$, nine blocks of two or more nodes and five isolated nodes (Table 2, Figure 4). Four qualify and are separable: (1) outdoor dust and occupational exposure, construction sites with their workers and neighbours, street vendors and migrant labour, 17 nodes, $\varepsilon_{\text{sym}} = 0.375$, and the lowest dependence on outside inputs ($\varepsilon_{\text{in}} = 0.125$), because dust at a fenceline or smoke at a stall is made locally. (2) The school gate and idling, 20 nodes, 0.556, with six actions, the most of any block. (3) Child and commuter time-activity and dose, 19 nodes, 0.823. (4) Ambient PM with near-road residents and delivery riders and the city’s health burden, 23 nodes, 0.934. The energy and traffic blocks are the tightest subsystems in the graph but hold no exposure or effect node, and the classroom CO₂ chain has no action inside it. Blocks 2 and 3 are the two halves of the school system, sources and levers on one side, the child’s dose and effects on the other, joined by gate concentration to child time at the gate (S20 to X04) and classroom exposure to child dose (X10 to X06).

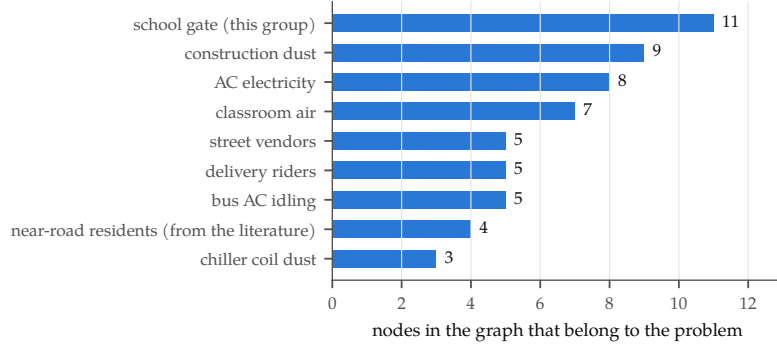


Figure 2: Coverage: nodes in the graph that belong to each problem raised in class (fixed node sets, Appendix A). Every candidate problem is represented with three to eleven dedicated nodes, the condition under which a decomposition can be compared across them.

Table 2: The nine blocks of two or more nodes. A block qualifies with an action and an outcome inside it.

rank	block	qualifies	n	W_{in}	ϵ_{sym}	ϵ_{out}	ϵ_{in}	$\epsilon_{transfer}$
1	outdoor dust and occupational (construction, vendors)	yes	17	56	0.375	0.250	0.125	0.214
2	school gate and idling	yes	20	63	0.556	0.365	0.190	0.143
3	child and commuter time-activity and dose	yes	19	62	0.823	0.290	0.532	0.081
4	ambient PM, near-road residents, riders, health burden	yes	23	91	0.934	0.363	0.571	0.066
5	dust, PM ₁₀ , chiller-coil fouling	yes	16	60	1.350	0.633	0.717	0.350
6	two-node fragment	no	2	1	0.000	0.000	0.000	0.000
7	energy, cooling and industry	no	26	89	0.258	0.124	0.135	0.101
8	traffic emissions, NO _x , ozone, fleet levers	no	28	104	0.548	0.423	0.125	0.298
9	classroom CO ₂ , ventilation, cognition, absenteeism	no	4	12	0.917	0.083	0.833	0.000

Stability (Figure 5). *Resolution*: the school block is identical at 0.8 and 1.0 and reappears at Jaccard 0.70 (Jaccard 1912) at 1.2 and 1.5, the near-road block is identical at 1.0 and 1.2 and reappears at 0.87 at 1.5, and at 0.6 block 1 is unchanged while blocks 2 and 3 merge. *Algorithm*: Leiden (Traag et al. 2019) returns the school block at Jaccard 0.85 and Clauset-Newman-Moore greedy modularity (Clauset et al. 2004) at 0.95. *Weights*: with every ordinal weight bumped one level at random 200 times, the school block reappears at Jaccard 0.6 or better in 66 % of trials (mean 0.668), the other three in 29 to 30 % (mean 0.53 to 0.54). The rank order is less stable than the membership: block 1 leaks least in 40 % of trials, block 2 in 10 %, and the five-node street-vendor chain, a fragment of block 1, in 34 %. *Seeds*: 50 Louvain seeds return the blocks at mean Jaccard 0.63 to 0.78.

What the step delivers. A menu of four separable problems with a lever and an outcome inside each, and a measured stability for it. Leakage is a ratio, so a small closed chain scores well however little it matters, which is why the vendor chain wins a third of the perturbed reruns. Ranking the menu needs a criterion stated outside the graph, which is step 3.

4 Step 3: exposure load ranks the menu

Ott (1982) defined a person’s exposure as the integral of concentration over time in contact with it, and Duan (1982) showed it can be computed piecewise over microenvironments. Summed over the people in a microenvironment,

$$L = N \times h \times \Delta C, \quad (2)$$

with N the people in it, h their hours per day there and ΔC the PM_{2.5} increment the block’s own sources add, in person $\mu\text{g}/\text{m}^3 \text{ h}$ per day, reported in millions. PM_{2.5} is the one pollutant every block has a fetched increment for. The metric measures where a block’s own pollution meets the most breathing. It does not weight susceptibility or toxicity per microgram (Section 8). Intake fraction (Bennett et al. 2002) would need an emission rate per source, which no block has for Abu Dhabi.

Inputs (Table 3). Populations are Abu Dhabi statistics: 396,230 pupils in 2022 (Abu Dhabi Chamber of Commerce and Industry 2024), 2,762,715 employed persons at 30 September 2024 (Statistics Centre Abu Dhabi 2024) of whom 14.1 % work in construction (Statistics Centre Abu Dhabi 2020), so $2,762,715 \times 0.141 = 389,543$, of whom 83.4 % work outdoors after Al Hurini et al. (2024) (331 of 397 labour-camp workers outdoors over two hours a day), so $389,543 \times 0.834 = 324,879$. School transport carries 237,111 pupils on 8,568 buses

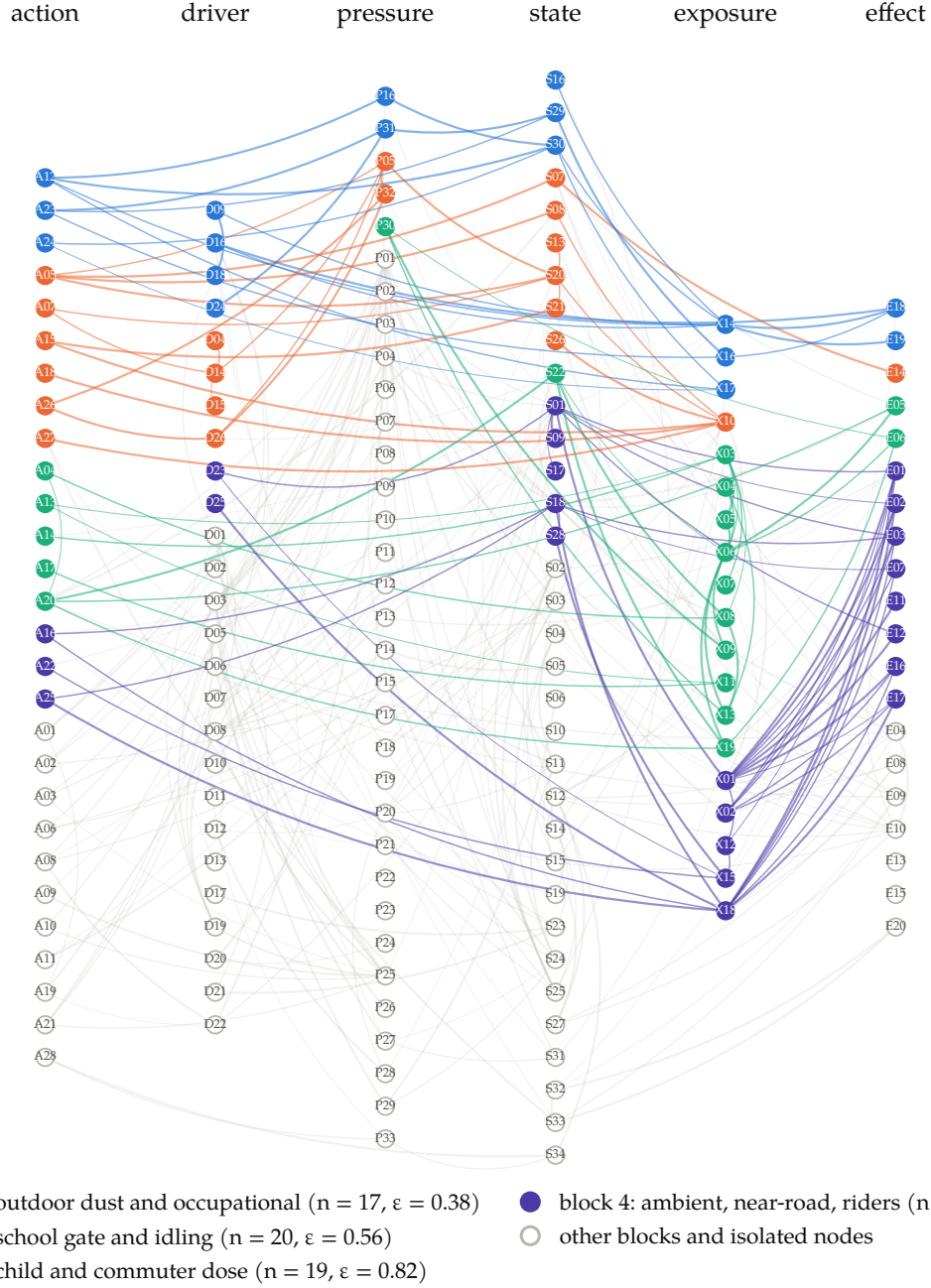


Figure 3: The 160-node implication graph by DPSEEA layer, every edge a cited influence (line weight follows the ordinal weight). Nodes and internal edges of the four separable blocks of Section 3 are coloured: construction sites, their workers and neighbours and the street vendors (blue), the school gate and vehicle idling (orange), the child's time-activity and dose (green), ambient PM with near-road residents, delivery riders and the city's health burden (violet).

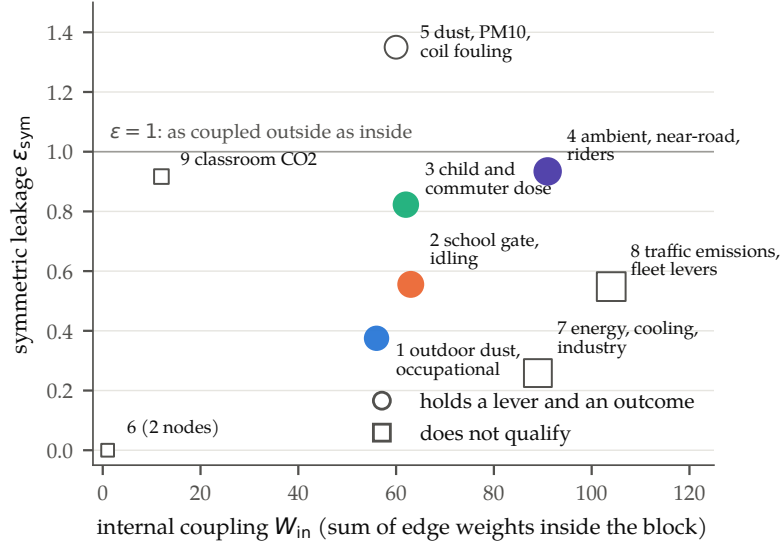


Figure 4: Symmetric leakage against internal coupling. Circles qualify, squares do not, marker area follows node count. Everything below $\varepsilon = 1$ is more coupled inside than across its boundary. The four coloured blocks are the menu.

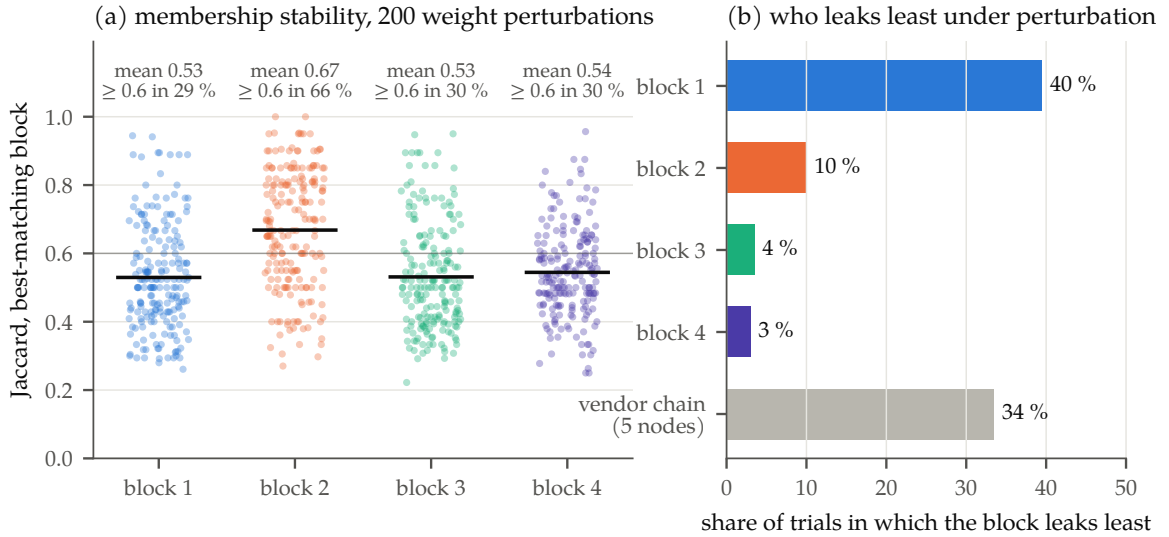


Figure 5: Stability under 200 random one-level perturbations of every edge weight. (a) Jaccard similarity between each block and its best-matching block in the perturbed partition, one dot per trial, bar at the mean. (b) Share of trials in which each block is the lowest-leakage qualifying block.

run by 206 operators at 672 schools and nurseries (Abu Dhabi Mobility 2024a), 59.8 % of the roll, with the car-to-walk split of the remainder from Badri et al. (2012). Delivery riders are the 26,186 commercial motorcycles (Abu Dhabi Mobility 2024b). The 2024 population is 4,135,985 with 68.2 % in the Abu Dhabi Region (Statistics Centre Abu Dhabi 2025b). Increments come from measured studies elsewhere. Bus cabin: in-cabin $\text{PM}_{2.5}$ 20 ± 18 against ambient 7 ± 5 and roadway $13 \pm 12 \mu\text{g}/\text{m}^3$ over 597 trips on 188 buses, mean trip 40 minutes (Adar et al. 2015). School gate: a mean drop-off increment of 3.45 across 86 schools (Adams and Requia 2017), 4.11 measured at the most-bused of four schools (Ryan et al. 2013). Construction: fine-fraction increment of 5 to 8 (PM_{10} about 110 to 125) from a demolition site’s fixed and mobile measurements against background (Azarmi and Kumar 2016), diesel elemental carbon 3.4 against 0.9 on site (Ferree et al. 2024). Near-road: 31 US near-road $\text{PM}_{2.5}$ monitors run 6 to 10 % above average, 0.6 to $1.1 \mu\text{g}/\text{m}^3$ (Seagram et al. 2019), the zone of influence for PM mass 100 to 400 m (Zhou and Levy 2007), and many studies find no PM-mass gradient at all (Karner et al. 2010), so this is an upper end. Indoor-to-outdoor 0.4 to 0.6 in sealed Gulf homes (Yuan et al. 2020), motorcycle at four times a car (Patel et al. 2016), time at home from the US activity survey (Klepeis et al. 2001), the charcoal-cart microenvironment of $196 \mu\text{g}/\text{m}^3$ from New York (Nahar et al. 2020).

Result (Figure 6). Construction workers $16.9 \text{ Mperson } \mu\text{g}/\text{m}^3 \text{ h/day}$ (13.0 to 20.8), near-road residents and riders 7.0 (3.1 to 18.2), the child commute 4.47 (4.3 to 4.7) of which the bus term alone is 4.11, the school gate 0.68 (0.34 to 1.22). Per person per day: a construction worker $52 \mu\text{g}/\text{m}^3 \text{ h}$, a delivery rider 48, a bus pupil

Table 3: The exposure-load terms, $L = N \times h \times \Delta C / 10^6$ in M person $\mu\text{g}/\text{m}^3 \text{ h}/\text{day}$. Ranges follow the stated low and high values of each assumed input. Rows in italics could not be computed for want of a count. [†] $N = 2.8 \text{ M} \times f$ with $f = 0.3$ (0.2 to 0.5) the unsourced fraction of the Abu Dhabi Region living within 300 m of an arterial, and $\Delta C = 0.85$ (0.6 to 1.1) $\times$ an indoor-to-outdoor ratio of 0.5 (0.4 to 0.6). The riders' high bound uses the roadway concentration of 13. The gate's high bound uses the measured 4.11.

block	microenvironment	N	h (h/day)	ΔC ($\mu\text{g}/\text{m}^3$)	L
1	construction workers on site	324,879	8	6.5 (5 to 8)	16.9 (13.0 to 20.8)
1	<i>site-neighbour residents</i>	<i>no count</i>	16	<i>none extracted at 100 m</i>	<i>not computed</i>
1	<i>vendors at charcoal or gas stalls</i>	<i>no count</i>	10	152	1.5 per 1,000 vendors
2	pupils in the gate zone	396,230	0.5 (0.25 to 0.75)	3.45 (3.45 to 4.11)	0.68 (0.34 to 1.22)
3	pupils on school buses	237,111	1.33	13	4.11
3	pupils driven by car	115,676	0.67	3	0.23
3	pupils walking	43,443	0.5	6	0.13
3	total				4.47 (4.3 to 4.7)
4	delivery riders on the road	26,186	8 (6 to 10)	6 (6 to 13)	1.26 (0.94 to 3.40)
4	near-road residents at home [†]	840,000	16	0.43	5.7 (2.2 to 14.8)
4	total				7.0 (3.1 to 18.2)

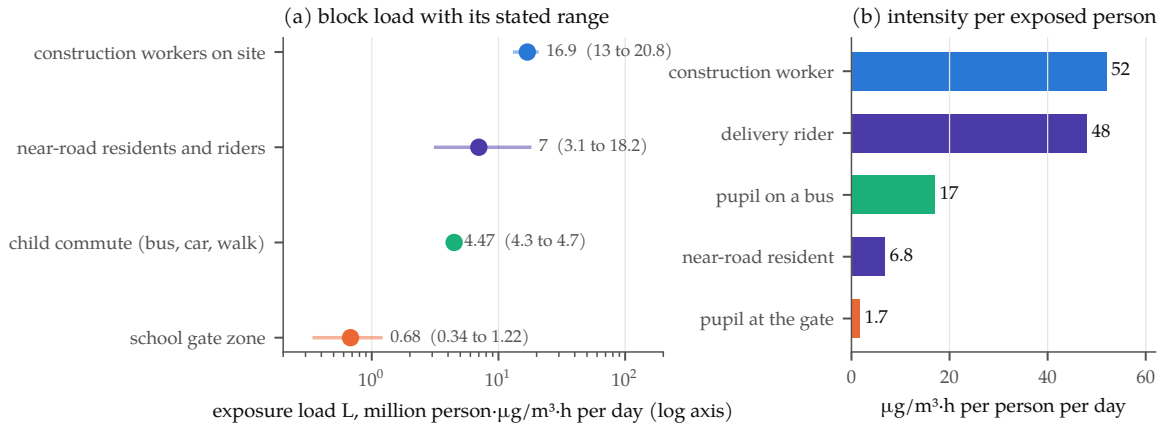


Figure 6: (a) Exposure load per block on a log axis with the range implied by the stated low and high inputs. (b) The same arithmetic per exposed person.

17, a near-road resident 6.8, a pupil at the gate 1.7. In PM_{10} the construction row is 299 M against an Abu Dhabi Region ambient of $114.6 \mu\text{g}/\text{m}^3$ (Statistics Centre Abu Dhabi 2025a), so a site roughly doubles what a worker breathes. By impact the menu picks outdoor construction labour, then near-road residents. What a three-student group can reach, change and measure within a semester is a set of criteria of its own, and step 4 states them and applies them to every candidate intervention rather than to the problem in the abstract.

5 Step 4: generating and screening the interventions

Five problems. Block 1 holds two receptor groups with different sources and is split, so that each need statement has one microenvironment (Table 4).

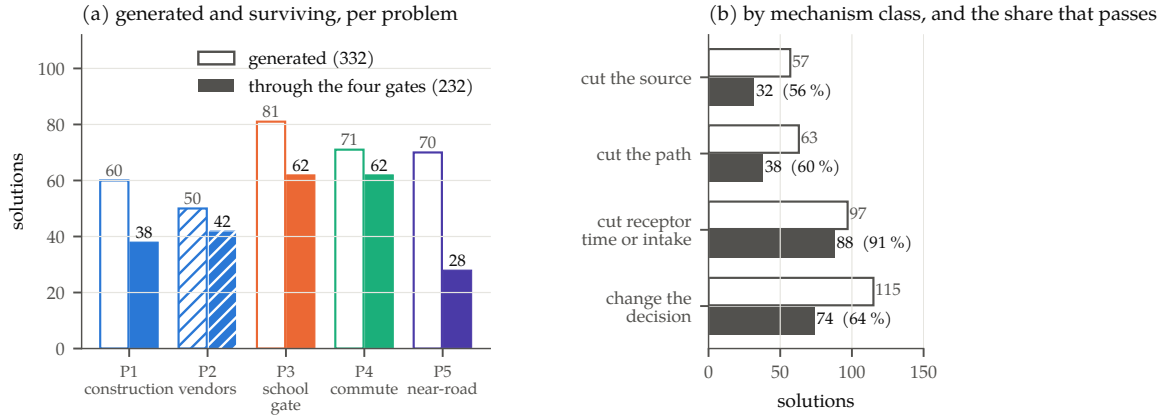
Generation by morphological analysis on the edges. Every internal edge of a problem block is a documented mechanism and therefore a place to intervene. Following Zwicky (1967) as formalised by Ritchey (2006), the generation space is the product of the block's internal edges and four mechanism classes, and every cell was filled with at least one concrete intervention. The classes are the DPSEEA level at which the edge is cut: *cut the source* (the pressure level), *cut the path* (the state level), *cut receptor time or intake* (the exposure level), and *change the decision* (a rule, a price, information or enforcement, where the deeper leverage points of Meadows (2012) sit). The blocks' 25, 16, 7, 29 and 42 internal edges yield 81, 60, 50, 71 and 70 interventions, 332 in all (Figure 7). Each row carries its edge, so the mind map the brief asks for is the graph itself.

Gates. Four binary gates, the conjunctive form of a hard constraint (Belton and Stewart 2002), applied to all 332 before any scoring: **G1 reach**, the group can get to the user or the site this term, **G2 semester**, an effect can be produced and measured by December with instruments the group can borrow or buy, **G3 additive**, not already the standing rule or product in Abu Dhabi, **G4 no law change**. 232 pass. The near-road block loses most (28 of 70) because setbacks, zoning, fleet procurement and corridor measures need a regulator. Failed rows stay in the inventory with the gate that failed them.

Rubric. Survivors are scored by a weighted-sum model (Triantaphyllou 2000), the additive form appropri-

Table 4: The five problems, each with its inclusion boundary, its exclusion boundary and its load.

problem	in	out	L
P1 construction workers and site neighbours	the dust source, the fenceline, the two exposed groups	vendors, traffic, dust storms	16.9
P2 street food vendors	fuel, stall, concentration, vendor	customers, kitchens, traffic	no count
P3 school gate and idling	car and bus idling, gate concentration, infiltration, classroom	the commute, home, ambient dust	0.68
P4 child commute and daily dose	time-activity, in-vehicle air, breathing rate, peaks, lung and cognition	the gate, the sources	4.47
P5 near-road residents and delivery riders	the near-road gradient, on-road air, the two groups, their out-comes	schools, sites, stalls, regional dust	7.0

**Figure 7:** The funnel. (a) Interventions generated on each problem's edges and the number that pass the four gates (P2 hatched, it shares block 1 with P1). (b) The same by mechanism class. Receptor-side measures pass most often because they need nobody's permission, which is why the rubric carries a root-cause criterion.

ate when the criteria are preferentially independent (Keeney and Raiffa 1993), on eleven criteria each scored 1 to 5 against written anchors with stated integer weights, maximum 90 (Table 5). Five criteria are the instructor's families. Six are ours: impact, anchored on the loads of step 3, measurability, time to effect, and the three that separate a cure from a shield, *novelty*, *cure* and *transfer*, the last being the graph's own $\epsilon_{\text{transfer}}$ at the level of one intervention. The problem-level scores are child commute 70, vendors 69, construction 67, school gate 59, near-road 59. Among the 232 interventions the leaders are the per-bus cabin air service (79), a clean cooking zone pilot at one market (74), induction grills at stalls (73), LPG conversion (72), exhaust-leak inspection of every bus (71) and engine-off training for hired family drivers (71) (Figure 8a).

Criteria ablation (Figure 8b, Figure 9b). Removing *novelty*, *cure* and *transfer* leaves a rubric on which the cabin service and a bedroom air purifier tie at 58 of 65 and purifiers hold three of the top eight ranks. With the three criteria in, the purifiers fall to ranks 64 and beyond and the source-side rows lead by 15 points at the stated weights. The three criteria are what let the rubric measure what the brief asks for, a cure rather than a shield.

Robustness to the weights. Stochastic multicriteria acceptability analysis (Lahdelma and Salminen 2001, Lahdelma et al. 1998) draws weight vectors from a distribution expressing what is known about the weights, ranks the alternatives under each draw, and reports the rank acceptability index b_i^r , the share of draws in which alternative i holds rank r (Butler et al. 1997, Tervonen and Lahdelma 2007). Three distributions, 10,000 draws each, seed 0: *centred*, a Dirichlet with concentration equal to the stated weights, $\pm 50\%$, each weight uniform on half to one-and-a-half times its value, and *uniform*, a flat Dirichlet on the simplex, SMAA with no preference information at all. The cabin service holds rank 1 in 90.3 % of centred draws, 100 % of $\pm 50\%$ draws and 86.0 % of uniform draws (Figure 9a). No other row exceeds 6.3 % at rank 1 under any scheme. Dropping any one criterion leaves it first in all eleven cases. The pick depends on the criteria, not on the weights.

Table 5: The eleven criteria, their weights and the anchors of the 5 and 1 levels.

criterion	weight	5	1
user	1	the exposed person asks for it	imposed with no felt benefit
feasibility	2	built or done with our resources this term	needs an organisation we are not
implementability	2	one person must say yes	a regulator plus an industry
usability	1	passive, no behaviour change	needs daily discipline
market and business	1	clear buyer, repeatable	none
impact	3	removes most of the block’s load for everyone in the microenvironment	marginal
measurability	2	a before-and-after reading at one point	only a survey
time to effect	1	the day it is installed	years
novelty	2	nothing does this, a startup could own it	commodity product or standard rule
cure	2	removes the emission or the behaviour for everyone in the microenvironment	shields one receptor, source untouched
transfer	1	nothing moved elsewhere	burden moved to another group and to energy or waste

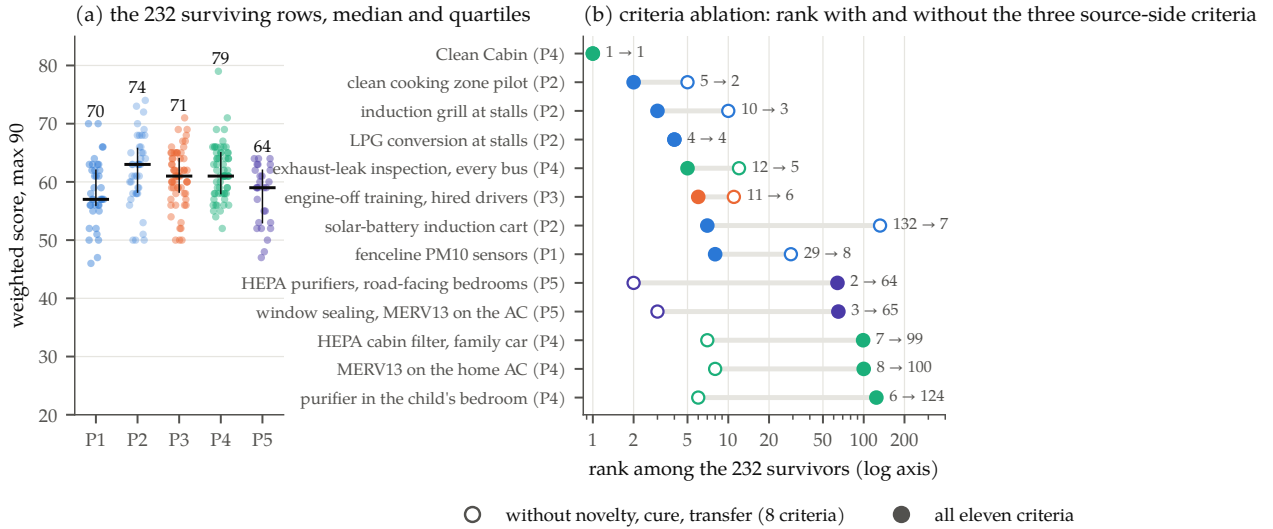


Figure 8: (a) Weighted scores of the 232 surviving interventions by problem, medians and quartiles, top score labelled. (b) Rank of the leading rows without (open) and with (filled) the three source-side criteria, on a log axis.

6 Step 5: failure-mode analysis fixes the design specification

The leading design on the school-gate block after scoring, gate orchestration (arrival slots and engine-off staging for buses, a just-in-time call for cars, a rolling lane, a dashboard, a gate sensor, 79 of 90), was subjected to a premortem (Klein 2008): assume it has been piloted and has failed, and explain why, a method that rests on the prospective-hindsight finding of Mitchell et al. (1989). To make the output examinable it was structured as a failure modes and effects analysis (International Electrotechnical Commission 2018): modes enumerated layer by layer (heat, traffic mechanics, behaviour, institutions, the instrument, the mechanism, the market, the course), each given a severity from a definition fixed in advance (*fatal*, the design as specified does not deliver its claim, *major*, a large part of the effect or the pilot is lost, *minor*, friction), and each tied to a graph edge or a fetched source. Twenty-nine modes, 5 fatal, 18 major, 6 minor (Figure 10). Four of the fatal modes are mechanisms rather than execution risks (Table 6), and the fifth is that an app releasing children on a parent’s position handles minors’ location data, which no school signs with a student group as data controller within a term. Two more cut the design’s claims: the car half exists as a product, PickupLine (iSchoolRide 2026), and the classroom carry-over measured in a naturally ventilated UK classroom (Kumar et al. 2020) and in Barcelona (Rivas et al. 2014) mostly does not apply to sealed Gulf rooms with indoor-to-outdoor ratios of 0.2 to 0.6 (Yuan et al. 2020). Scored with the modes in, gate orchestration is 57.

The specification. The largest exposure term in the school system is the bus cabin, 4.11 of the commute

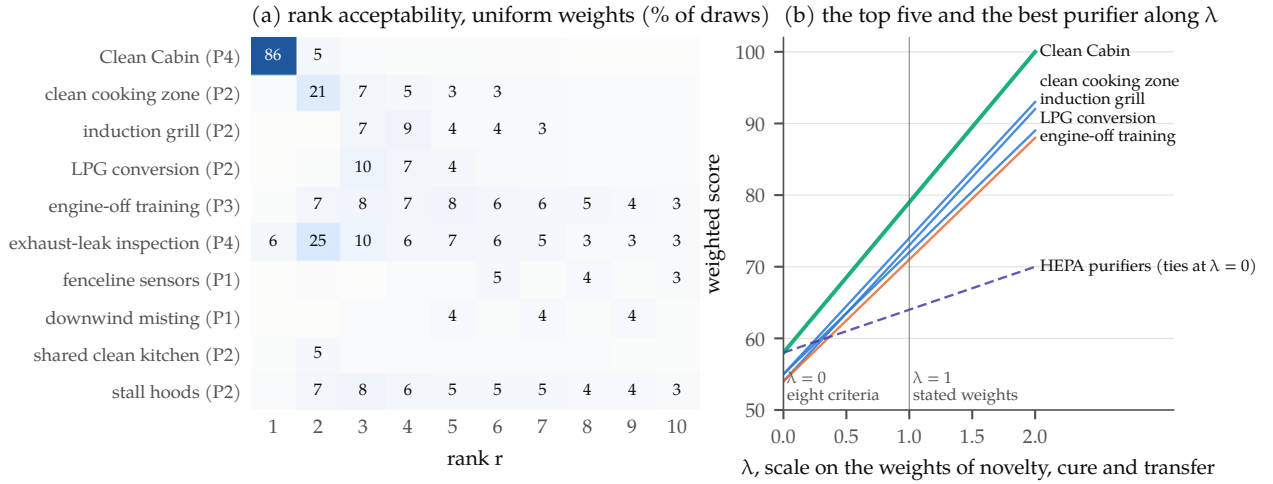


Figure 9: (a) Rank acceptability of the ten leading rows under uniform weights, the share of 10,000 draws in which each row holds each rank. (b) Weighted score of the top five and of the best purifier as the weights of novelty, cure and transfer are scaled together by λ .

Table 6: The four fatal mechanisms of a gate-side design, and the design rule each one sets.

mode	mechanism and evidence	rule for the design
F1 heat	in Abu Dhabi the idle is the air conditioning: a bus staged engine-off in September is a 45 °C cabin in minutes, and the idle-reduction evidence (Alternative Fuels Data Center, US Department of Energy 2015) is a cold-climate cab-comfort document	never ask anyone to sit in the heat with the engine off
F2 relocation	a just-in-time call moves the wait, drivers and nannies are already parked nearby with the engine running (Emirates 24 7 2026), the load moves to the next street	remove the emission, not the place it happens
F5 the morning	the call exists only at dismissal, the morning is a throughput queue against a fixed bell and the threefold peak of Kumar et al. (2020) was a drop-off peak	act on both sessions of the day
F19 instrument	the gate increment is 3 to 4 $\mu\text{g}/\text{m}^3$ (Adams and Requia 2017, Ryan et al. 2013), background explained 80.9 % of the variance at a New York dismissal (Richmond-Bryant et al. 2009), the Abu Dhabi background is 44 $\mu\text{g}/\text{m}^3$ and dust-dominated (Health Effects Institute 2022), low-cost optical sensors need site calibration and respond to humidity and composition (Zheng et al. 2018) and do not see ultrafines	measure inside a closed volume, as a ratio that cancels the background

block's 4.47 M person $\mu\text{g}/\text{m}^3 \text{ h/day}$, six times the gate, and in-cabin concentrations are largely the bus's own doing, highest when idling with no other diesel vehicle in front (Behrentz et al. 2005). With the rules of Table 6 the specification reads: act on the block's largest load term, at the source, inside a closed volume where the effect is measurable as a ratio, without engine-off in the heat, and without any parent, driver or app in the loop. Clean Cabin is that specification. On the unchanged rubric it scores 79, the top row alone (Figure 12). Of the 29 modes, 18 do not apply to it, four apply and are designed into the pilot (F9 restart pulses, F17 one operator's consent, F26 the buyer, F28 a three-student team), three concern the gate lane, bus slots and marshals and have no counterpart, and four are open (F4 dust days, F18 procurement and liability, F27 defensibility, F29 the instructor's question).

7 Clean Cabin: a per-bus cabin air service

Design (Figure 11). *Diagnose*: two low-cost PM and CO_2 sensors per bus, one in the cabin and one on the roof, co-located and cross-calibrated before installation, logging at idle and in motion. The metric is the cabin-to-roof ratio, so the dust background and the humidity and composition response the matched sensors share (Zheng et al. 2018) cancel, the expected signal is large (cabin 20 against ambient 7 $\mu\text{g}/\text{m}^3$ in Adar et al. 2015), and the comparison is within-bus, before and after, a paired design. *Fix at the source*: for the buses that fail,

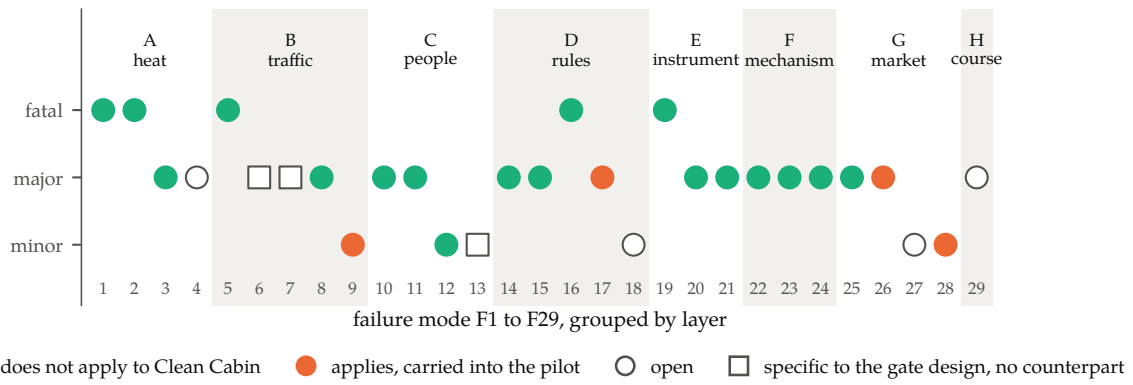


Figure 10: The 29 failure modes of a gate-side design by layer and severity, and whether each applies to Clean Cabin. None of the fatal modes does. Four apply and are carried into the pilot: the operator's consent, the buyer, restart pulses and the size of the team.

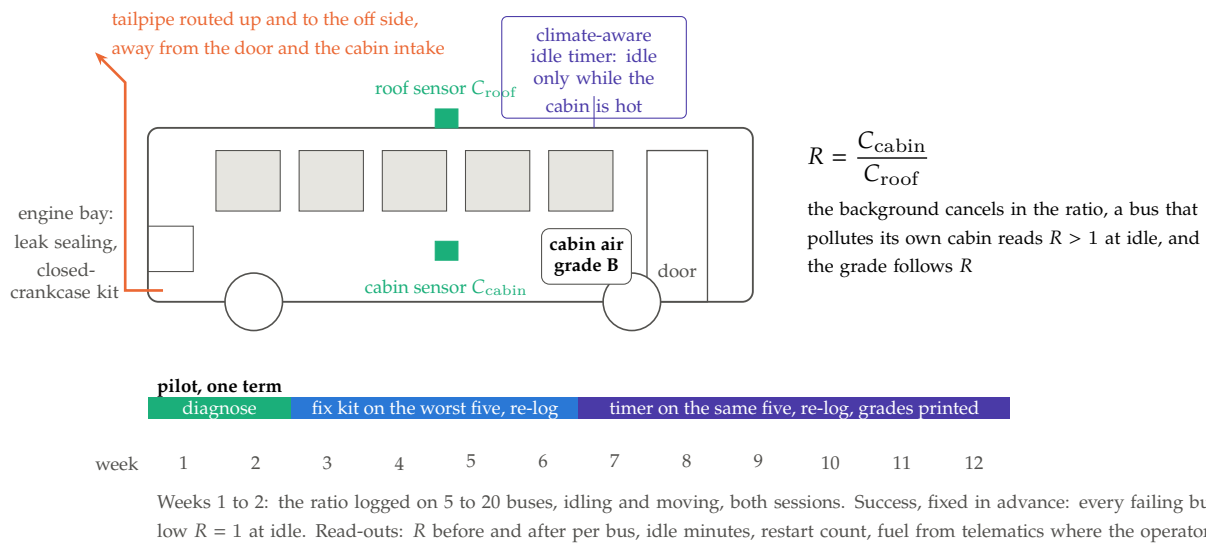


Figure 11: The four parts of the service on one bus, the metric, and the pilot.

exhaust-leak sealing, a closed-crankcase ventilation kit and the tailpipe routed upward and to the off side. No filter, no consumable, nothing moved elsewhere. *Climate-aware idle timer*: the engine idles only while the cabin is above a set temperature. *Rate and publish*: each bus receives a cabin air grade from its ratio, printed on the bus and shown to the school and to parents, which is what parents choose on, what a school writes into its transport contract, and the business.

Mechanism and prior art (Figure 13). Dual-tracer measurements attribute about 50 % of in-cabin $PM_{2.5}$ to the bus's own emissions with windows closed, over three quarters of that from the crankcase (Ireson et al. 2011), and the ride is a tenth of a child's day and a third of its black carbon exposure (Behrentz et al. 2005), with in-bus black carbon of 0.9 to $19 \mu g/m^3$ and NO_2 of 64 to $220 \mu g/m^3$ (Sabin et al. 2005). The kit is proven in the cabin: on a test track a closed-crankcase system cut in-cabin ultrafine counts by 50 to over 100 %, a tailpipe filter without it did not, and a well-maintained inspected bus ran $2.7 \mu g/m^3$ in the cabin (Martinez-Morett et al. 2008). On route, crankcase filters with an oxidation catalyst cut in-cabin hopanes, PAH and aliphatics by 37, 50 and 43 % (Trenbath et al. 2009), tests in three states found 50 to 60 % lower cabin particles (Pedde et al. 2023), and across 188 buses cleaner fuel or retrofits ran 10 to 50 % lower on board with respiratory outcomes to match (Adar et al. 2015). One study found no unequivocal cabin decrease despite tailpipe cuts of 20 to 94 % (Zhang and Zhu 2011), which is why every bus is diagnosed before and after. Outcomes follow: 2,656 retrofitted Georgia buses raised respiratory fitness and English scores (Austin et al. 2019), and a randomised rebate programme raised attendance (Pedde et al. 2023). Every prior instance was a government retrofit or rebate programme delivered fleet-wide. A per-bus diagnosis by cabin-to-roof ratio, a published per-bus grade, a timer keyed to cabin temperature and any measurement of a school-bus cabin in the UAE: none was found. Novelty 4 rests on those four, and the physics being twenty years old is what lets the pilot promise a measured result in weeks.

Scale and market. 237,111 children on 8,568 buses run by 206 operators for 672 schools and nurseries,

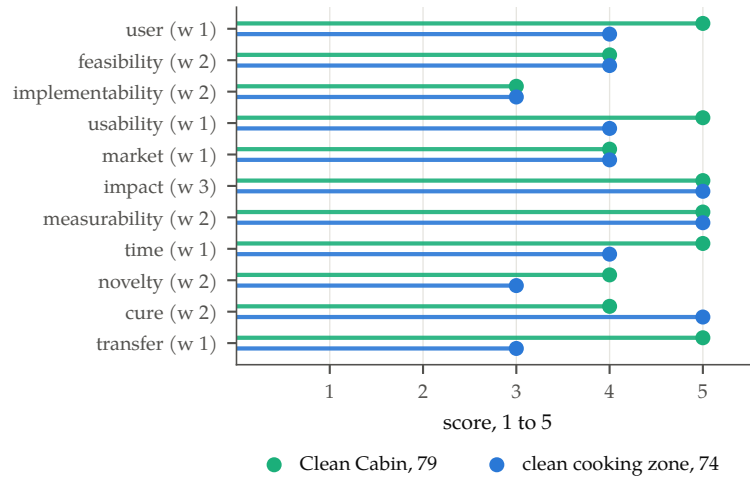


Figure 12: The eleven criterion scores of Clean Cabin against the runner-up. It leads or ties on nine criteria and trails only on cure, where replacing a fuel removes the emission outright.

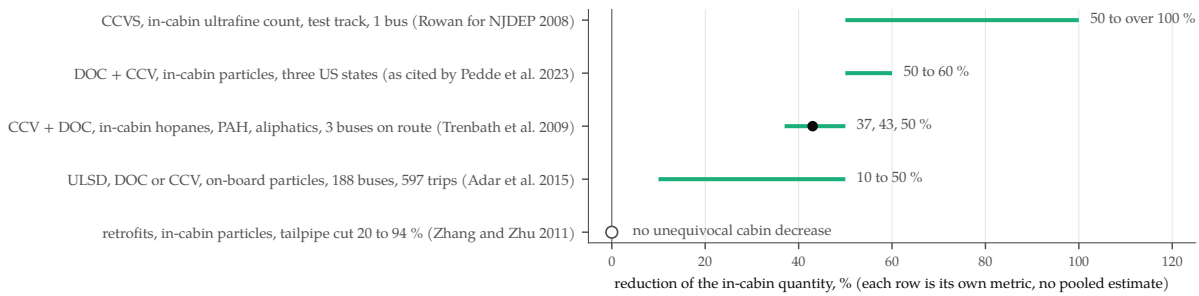


Figure 13: Measured in-cabin reductions from the source-side kit, one row per study with its own metric and sample. No pooled estimate is drawn because the quantities differ.

80 minutes a day at $13 \mu\text{g}/\text{m}^3$ above ambient, and a regulator whose transport policy fixes journey length, supervision and who may collect a child (Khaleej Times 2025) but is silent on the air in the cabin. Diagnosis and grade are sold per bus, the kit is fitted through the operator's workshop, the re-check is termly. Operators buy the grade parents ask for, schools buy a contract clause they can verify, and the regulator can adopt the grade as a standard later without the pilot needing it. The first partner is the university's own shuttle service, to be confirmed, then one operator's twenty buses at one school.

8 What is assumed, transported or unmeasured

Every increment in Table 3 is transported from another city and no UAE measurement exists for any of them, which is the first measurement the pilot makes. The near-road load rests on an unsourced fraction of residents within 300 m of an arterial, which a GIS overlay would settle. Time at the gate, the delivery shift and the car commute are stated assumptions, and the 8-hour day is the statutory figure. The load ignores susceptibility (Gauderman et al. 2004, 2007) and toxicity per microgram, and a child weight of two or three does not change the ranking. Leakage is size-blind and modularity has a resolution parameter, which is why the decomposition is reported as a menu with its stability. The eleven scores were assigned by the authors against written anchors and no inter-rater reliability was measured. SMAA covers uncertainty in the weights, not in the scores. The failure-mode list is what three people enumerated in a session, and the seven modes marked open or not applicable were classified, not analysed. The fuel and CO_2 of idling for cooling are outside the design.

9 Conclusion

A cited graph decomposed by near-decomposability turns a system into a menu of four separable problems and measures how stable the menu is. An exposure load with every input sourced ranks it. Morphological generation on the graph's edges produces 332 interventions that can be enumerated rather than remembered. A gated, weighted, stochastically stress-tested rubric ranks them, and a failure-mode analysis fixes the spec-

ification the winner has to meet. The result is a per-bus cabin air service for the 237,111 children who ride Abu Dhabi's school buses, built from twenty-year-old evidence that nobody has applied here, holding rank 1 in 86 % of rankings drawn with no information about our weights, and measurable inside a closed volume with instruments a student group can afford. The next step is one operator's twenty buses and a sensor in each.

A Reproducibility

Two folders hold everything. `agrl130-graph/`: `nodes.csv`, `eight edges-lane*.csv` with matching `sources-lane*.md`, `decompose.py` (build, leakage, Louvain, resolution sweep, Leiden and greedy cross-checks, sensitivity, the problem node sets of Figure 2), `perturb.py` (the 200 trials per block), `lanes/anchor_test.py`. `agrl130-solutions/`: `inventory.py` (332 interventions with edge, class, gates and scores), `extra.py` (novelty, cure, transfer), `build.py` (writes `solutions.csv` and the workbook), `robustness.py` (SMAA, leave-one-out, the λ sweep). `agrl130-report/figures` draws every figure from those files, `refs.py` resolved every reference at Crossref and `bib.py` wrote the bibliography from the returned records.

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